

SIMATS ENGINEERING
Department of Computer Science and Engineering
ASSESSMENT TOOL 3 – CODING CHALLENGE QUESTIONS

Course Code:	CSA12	Course Title:	Computer Architecture
CO Assessed:	CO1 – Precision (BL4)	Assessment:	Coding challenge questions
Weightage:	40%	Total Marks:	100
CO1 Statement:	Analyse the functional units of a computer system including CPU, memory, bus structures, and I/O components by examining instruction execution cycles, addressing modes, and performance metrics such as CPI, clock cycles, and benchmarking		
Student Name:	Kolar Umme Sadiya	Reg. No.:	192425117
Section / Batch:	AI&ML/3 rd year	Date:	30-07-2026

Instructions to Students

- Analyze the given scenario and identify the problem requirements before developing the solution.
- Design the algorithm or flowchart and select the appropriate 8085 instructions, registers, and addressing modes required for implementation.
- Develop and execute the 8085 Assembly Language Program (ALP) using a simulator or trainer kit to solve the given problem.
- Verify and validate the output by testing the program with the given input data and ensuring the expected results are obtained.
- Interpret the program execution by explaining the logic used, the role of registers, memory locations, flags, and the final output generated.

QUESTIONS

1. A warehouse stores the quantity of products available in eight storage racks. Before dispatch, the warehouse manager wants the processor to arrange the quantities in ascending order and identify the rack with the lowest stock for replenishment. Write an 8085 ALP to
 - Sort the quantities in ascending order.
 - Store the sorted values.
 - Identify and store the minimum quantity.
2. During data transmission, a block of eight bytes is copied from one memory region to another. To ensure successful transmission, each byte in the destination memory must be compared with its corresponding source byte. Write an 8085 ALP to
 - Copy the data block.
 - Compare each source byte with the copied byte.
 - Count the number of mismatches.
 - Store the mismatch count.
3. A communication controller receives an 8-bit binary code. According to the communication protocol, the code is valid only if it contains an even number of 1s. Develop an 8085 ALP to
 - Count the number of set bits.
 - Determine whether the parity is even or odd.
 - Store a validity flag in memory.

Code of Conduct

I Kolar Umme Sadiya 192425117 (Name / Reg No) certify that this submission is my original work and that I have adhered to the guidelines specified for this assessment. I understand that any violation of academic integrity rules will result in disciplinary action.

Signature of the Student: Kolar Umme Sadiya

RUBRICS FOR CALCULATION

Criteria	Weightage	Level 4 – Exemplary (5 Marks)	Level 3 – Proficient (4 Marks)	Level 2 – Developing (2–3 Marks)	Level 1 – Beginning (0–1 Mark)
Problem Understanding	20%	Clearly understands problem constraints, edge cases, and competitive requirements	Understands problem with minor gaps in constraints	Partial understanding; misses key constraints	Misinterprets problem or constraints
Algorithm	25%	Selects optimal algorithm with best time and space complexity for competitive constraints	Uses efficient algorithm with minor scope for improvement	Uses basic/inefficient approach	No clear algorithmic approach
Code Implementation	20%	Writes correct, efficient, and clean code that passes all constraints	Code works with minor errors or inefficiencies	Partially correct code; fails for some cases	Code does not execute or gives incorrect results
Competitive Performance	20%	Solves problem within time limits and handles large inputs effectively	Solves within acceptable time with minor delays	Struggles with time limits or larger inputs	Unable to solve within time constraints
Test Case Handling	15%	Handles all test cases including edge and hidden cases	Handles most test cases	Misses important edge cases	Fails on multiple test cases

Marks Evaluation:

Q. No	Problem Understanding (20)	Algorithm (25)	Code Implementation (20)	Competitive Performance (20)	Test case handling (15)	Total (out of 100)
1	/ 4	/ 4	/ 4	/ 4	/ 4	
2	/ 4	/ 4	/ 4	/ 4	/ 4	
3	/ 4	/ 4	/ 4	/ 4	/ 4	
	/ 20	/ 25	/ 20	/ 20	/ 15	/ 100

Course Coordinator

Faculty Incharge

1. Sort in Ascending Order and Store Minimum :

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code is as follows:

```
1 LXI H,2500H
2 LXI D,2600H
3 MVI C,08H
4 MVI B,00H
5
6 LOOP: MOV A,M
7 STAX D
8 LDAX D
9 CMP H
10 JZ NEXT
11 INR B
12
13 NEXT: INX H
14 INX D
15 DCR C
16 JNZ LOOP
17
18 MOV A,B
19 STA 2700H
20
21 HLT
```

The memory dump shows the following data:

Address (Hex)	Address	Data
09C4	2500	11
09C5	2501	22
09C6	2502	33
09C7	2503	44
09C8	2504	55
09C9	2505	66
09CA	2506	77
09CB	2507	88
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message shows: 0 Program assembled successfully.

2. Copy 8 Bytes, Compare and Count Mismatches:

The screenshot shows the GNUSim8085 - 8085 Microprocessor Simulator interface. The assembly code is as follows:

```
1 LXI H,2500H
2 LXI D,2600H
3 MVI C,08H
4 MVI B,00H
5
6 LOOP: MOV A,M
7 STAX D
8 LDAX D
9 CMP H
10 JZ NEXT
11 INR B
12
13 NEXT: INX H
14 INX D
15 DCR C
16 JNZ LOOP
17
18 MOV A,B
19 STA 2700H
20
21 HLT
```

The memory dump shows the following data:

Address (Hex)	Address	Data
09C4	2500	11
09C5	2501	22
09C6	2502	33
09C7	2503	44
09C8	2504	55
09C9	2505	66
09CA	2506	77
09CB	2507	88
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

The assembler message shows: 0 Program assembled successfully.

3. Count Set Bits and Check Even/Odd Parity :

GNUSim8085 - 8085 Microprocessor Simulator

File Reset Assembler Debug Help

Registers

Register	Value
A	00
BC	00 00
DE	00 08
HL	25 08
PSW	00 00
PC	42 25
SP	FF FF
Int-Reg	00

Flag

Flag	Value
S	0
Z	1
AC	1
P	1
C	0

Decimal - Hex Conversion

Decimal: 0 Hex: 0

To Hex To Dec

I/O Ports

0 - + 00

Update Port Value

Memory

0 - + 00

Update Memory

Load me at

```
1 LDA 2500H
2 MVI C, 08H
3 MVI D, 00H
4
5 LOOP: RAR
6 JNC SKIP
7 INR D
8
9 SKIP: DCR C
10 JNZ LOOP
11
12 MOV A, D
13 STA 2600H
14
15 ANI 01H
16 JZ EVEN
17
18 MVI A, 01H
19 STA 2601H
20 HLT
21
22 EVEN: MVI A, 00H
23 STA 2601H
24 HLT
```

Start 2500 OK

Address (Hex)	Address	Data
09C4	2500	5
09C5	2501	25
09C6	2502	18
09C7	2503	40
09C8	2504	10
09C9	2505	30
09CA	2506	0
09CB	2507	0
09CC	2508	0
09CD	2509	0
09CE	2510	0
09CF	2511	0
09D0	2512	0
09D1	2513	0

Line No Assembler Message

0 Program assembled successfully

Simulator: Idle

30°C Partly cloudy

Search

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